MH2500 Probability and Introduction to Statistics

Slides for Week 7-1

Topics: Gamma distribution

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Gamma Distribution

We have learnt that the waiting time until the first phone call follows exponential distribution.

Generalisation: How about the waiting time T_n until the n-th phone call?

Let $F(t)$ be the CDF of T_n, then

$$F(t) = P(T_n < t) = P(\text{at least } n \text{ calls in } [0, t])$$

$$= \sum_{k=n}^{\infty} \frac{(\lambda t)^k e^{-\lambda t}}{k!} = 1 - \sum_{k=0}^{n-1} \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

hence the probability density function is

$$p(t) = F'(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}.$$

$$
\begin{aligned}
p(t) = F'(t) &= \left(1 - e^{-\lambda t} - \sum_{k=1}^{n-1} \frac{(\lambda t)^k e^{-\lambda t}}{k!}\right)' \\
&= \left(-e^{-\lambda t}\right)' - \sum_{k=1}^{n-1} \left(\frac{(\lambda t)^k}{k!} e^{-\lambda t}\right)' \\
&= \lambda e^{-\lambda t} - \sum_{k=1}^{n-1} \left(\frac{k \cdot (\lambda t)^{k-1} \cdot \lambda}{k!} e^{-\lambda t} + \frac{(\lambda t)^k}{k!} e^{-\lambda t} \cdot (-\lambda)\right) \\
&= \lambda e^{-\lambda t} - \sum_{k=1}^{n-1} \left(\frac{(\lambda t)^{k-1} \cdot \lambda}{(k-1)!} e^{-\lambda t} + \frac{(\lambda t)^k}{k!} e^{-\lambda t} \cdot (-\lambda)\right) \\
&= \lambda e^{-\lambda t} - \sum_{k=1}^{n-1} \left(\frac{\lambda^k t^{k-1}}{(k-1)!} e^{-\lambda t} - \frac{\lambda^{k+1} t^k}{k!} e^{-\lambda t}\right) \\
&= \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}.
\end{aligned}
$$

gamma density function

The **gamma function** is defined as

$$\Gamma(x) = \int_0^\infty u^{x-1} e^{-u} du, \qquad x > 0.$$

Check that (using integration by parts)

$$\Gamma(x) = (x-1)\Gamma(x-1).$$

Proof:

$$\begin{aligned}
\Gamma(x) &= \int_0^\infty e^{-u} u^{x-1} du \\
&= \left[-e^{-u} u^{x-1}\right]_0^\infty + \int_0^\infty \left(e^{-u} \cdot (x-1) u^{x-2}\right) du \\
&= (x-1) \int_0^\infty \left(e^{-u} \cdot u^{(x-1)-1}\right) du \\
&= (x-1)\Gamma(x-1).
\end{aligned}$$

Thus,

$$\Gamma(n) = (n-1)\Gamma(n-1) = \cdots = (n-1)\cdots 2\Gamma(1) = (n-1)!$$

by

$$\Gamma(1) = \int_0^\infty e^{-x}dx = 1.$$

The **gamma density function** depends on two parameters, $\alpha > 0$ and $\lambda > 0$ and is defined as

$$g(t) = \begin{cases} \dfrac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t}, & t > 0; \\ 0, & t < 0. \end{cases}$$

Properties of the Gamma distribution

Two parameters, α and λ

- When $\alpha = 1$, $\Gamma(1) = \int_0^\infty e^{-u} du = 1$ and the Gamma distribution reduces to $g(t) = \lambda e^{-\lambda t}$, exponential distribution
- The time T_n, the waiting time until the n-th event follows a Gamma distribution with parameters $\alpha = n$ and λ.
  (Gamma variable is the sum of n exponential variables)
- Applications: used to model arrival times in the Poisson process, e.g. time for failures in systems, physical quantities taking positive values such as rainfalls, and in finances such as size of insurance claims.

Example

Observed times separating a sequence of small earthquakes. Which is a better fit? (Recall that exponential model is memoryless.)

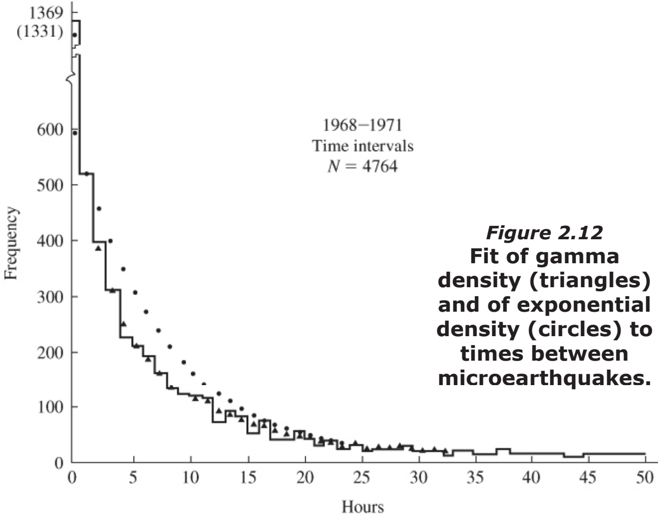

Figure 2.12
Fit of gamma density (triangles) and of exponential density (circles) to times between microearthquakes.

Expectation and Variance

Let $X \sim \Gamma(\alpha, \lambda)$. Then

$$\mathbb{E}(X) = \frac{\alpha}{\lambda}, \quad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2}.$$

Proof: (a)

$$\begin{aligned}
\mathbb{E}(X) &= \int_0^\infty x \cdot \left(\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \right) dx \\
&= \frac{1}{\Gamma(\alpha)} \int_0^\infty x \cdot \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx \\
&= \frac{1}{\lambda\Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{(\alpha+1)-1} dx \\
&= \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} \\
&= \frac{\alpha}{\lambda} \qquad \text{by } \Gamma(\alpha+1) = \alpha\Gamma(\alpha).
\end{aligned}$$

(b) We first find $\mathbb{E}(X^2)$:

$$\begin{aligned}
\mathbb{E}(X^2) &= \int_0^\infty x^2 \cdot \left(\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \right) dx \\
&= \frac{1}{\Gamma(\alpha)} \int_0^\infty x^2 \cdot \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx \\
&= \frac{1}{\lambda^2 \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{\alpha+1} dx \\
&= \frac{\Gamma(\alpha+2)}{\lambda^2 \Gamma(\alpha)} = \frac{(\alpha+1)\alpha\Gamma(\alpha)}{\lambda^2 \Gamma(\alpha)} \\
&= \frac{(\alpha+1)\alpha}{\lambda^2} \qquad \text{by } \Gamma(\alpha+2) = (\alpha+1)\alpha\Gamma(\alpha).
\end{aligned}$$

Thus,

$$Var(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{(\alpha+1)\alpha}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}.$$

Recall that $\Gamma(n, \lambda)$ is sum of n exponential random variables. Hence for $X \sim \Gamma(n, \lambda)$,

$$\mathbb{E}(X) = \frac{n}{\lambda}, \quad \mathrm{Var}(X) = \frac{n}{\lambda^2}$$

- The explanation that $\mathrm{Var}(X) = n \cdot \mathrm{Var}(Y)$ for $Y \sim \mathrm{Exp}(\lambda)$ will be revealed in Chapter 7.

Functions of a Random Variable

Suppose X is a random variable with density function $f(x)$ and Y is a random variable that can be expressed in terms of X, say, $\underline{\underline{Y = g(X)}}$ for some function g.

We would like to find the density function of Y.

1. Find the CDF of Y, i.e., $P\{Y \leq y\}$

   (Express in terms of probability of event about X).

2. Differentiate w.r.t. y

Example

Let U be a uniform random variable on $[0, 1]$ and let $V = 1/U$. Find the density of V.

Solution:

$$F_V(v) = P\{V \leq v\} = P\left\{\frac{1}{U} \leq v\right\} = P\left\{U \geq \frac{1}{v}\right\}$$

$$= 1 - \frac{1}{v} \qquad \text{(since U is a uniform random variable on [0,1].)}$$

By differentiating it, we obtain the density

$$f_V(v) = \frac{1}{v^2}, \qquad 1 \leq v < \infty.$$

Example

Find the density of $X = Z^2$ where $Z \sim N(0, 1)$.

Solution:

$$F_X(x) = P\{X \le x\} = P\{Z^2 \le x\} = P\{-\sqrt{x} \le Z \le \sqrt{x}\}$$
$$= \Phi(\sqrt{x}) - \Phi(-\sqrt{x}).$$

We find the density of X by differentiating the cdf. Recall $\Phi'(x)$ is the probability density function of the standard normal variable, and we denote it by $\phi(x)$. By the chain rule,

$$\begin{aligned} f_X(x) = (F_X(x))' &= \frac{d}{dx}(\Phi(\sqrt{x}) - \Phi(-\sqrt{x})) \\ &= \phi(\sqrt{x})\frac{d}{dx}(\sqrt{x}) - \phi(-\sqrt{x})\frac{d}{dx}(-\sqrt{x}) \\ &= \frac{1}{2\sqrt{x}}\phi(\sqrt{x}) + \frac{1}{2\sqrt{x}}\phi(-\sqrt{x}) \\ &= x^{-1/2}\phi(\sqrt{x}) \qquad \text{(by symmetry of } \phi.\text{)} \end{aligned}$$

From the density ϕ, we have

$$f_X(x) = \frac{x^{-1/2}}{\sqrt{2\pi}} e^{-x/2}, \qquad (x \geq 0).$$

This resembles gamma density with $\alpha = \lambda = 1/2$,

$$g(t) = \frac{t^{-1/2}}{\sqrt{2}\Gamma(\frac{1}{2})} e^{-t/2}, \qquad t \geq 0.$$

Since $f_X(x)$ and $g(t)$ only differ by at most a constant factor, and they both integrate to 1, they must be equal, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

This density is the **chi-squared density** with 1 degree of freedom. In general, when $\lambda = 1/2$ and $\alpha = n/2$, the gamma distribution is called χ_n^2 with n degrees of freedom.